Aman Kumar

ak2281198@gmail.com | Halo Mind Research | Personal Portfolio
LinkedIn | GitHub | Google Scholar

Research Profile

I work across computer vision and deep learning, with research spanning medical imaging, remote sensing, space-weather forecasting and degraded text recognition. My strengths are designing efficient neural architectures, building synthetic data pipelines when real labels are scarce, and testing whether models hold up under distribution shift through cross-domain evaluation and conformal prediction. I am a research intern at ISI Kolkata with Prof. Umapada Pal and at the University of Salford with Dr. Shivakumara Palaiahnakote, and I lead a student research team at SMIT. I am applying to Master's and Ph.D. programs for Fall 2027 to continue this work at a deeper level.

Education

Sikkim Manipal Institute of Technology (SMIT) Sikkim, India
Bachelor of Technology in Computer Science and Engineering (Data Science) *Jul. 2023 – Jul. 2027 (expected)*

Publications

Published & Accepted

- **A. Kumar**, L. Chhetri, D. Das. "SPECTRAFORGE: Domain-Equalized Frequency-Spatial Fusion for Synthetic Dermatology Detection." *Accepted for **Short Oral Presentation** at the IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 2026.
- L. Chhetri*, **A. Kumar***, D. Das, P. Ghosal. "Interpretable Solar Panel Defect Detection via Fuzzy Rule Extraction from Hierarchical Vision Models." *Accepted at the Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, 2026. ACM ICPS.
- L. Chhetri, **A. Kumar**. "DDV-GNet: High-Throughput Defect Detection for Space Manufacturing via Deep Delta Gated Networks." *Accepted for **Oral Presentation** at IEEE SPACE*, 2026.
- L. Chhetri, **A. Kumar**. "Optimizing Deep Learning for Brain Tumor Classification: A Comparative Ablation Study of Preprocessing and Augmentation Strategies." *Published at IEEE GCON*, 2026. DOI: 10.1109/GCON69192.2026.11649062
- A. Mukherjee, **A. Kumar**, S. Bala, S. R. Varma, A. Anand, H. Das, L. Chhetri. "Risk-Controlled Urban Change Detection: Conformal Prediction Wrappers for Provable Reliability in High-Resolution Satellite Imagery." *Accepted for **Oral Presentation** at ICCI*, 2026.
- R. Das, L. Chhetri, **A. Kumar**, P. Ghosal. "Intrinsic Neural Firewalls for Cyber-Physical Systems: Robust Anomaly Rejection via Deep Delta Residual Overwrites." *Accepted for **Oral Presentation** at WIN 6.0*, 2026.

Preprints

- **A. Kumar**, A. Anand, L. Chhetri. "Trust, but Verify: Epistemic Uncertainty-Gated Pseudo-Labeling for Safer Semi-Supervised Malaria Screening." *bioRxiv*, 2026. DOI: 10.64898/2026.10.13.758712
- **A. Kumar***, A. Anand*, L. Chhetri*. "DVARA: Frequency Gates for Adapting Frozen Underwater Image Enhancement Networks." *Preprints.org*, 2026. DOI: 10.20944/preprints202610.1474.v1

Conference Presentations

Short Oral Presentation, IEEE DSAA 2026 *Aerocity, New Delhi, India*
"SPECTRAFORGE: Domain-Equalized Frequency-Spatial Fusion for Synthetic Dermatology Detection."

Oral Presentation, IEEE SPACE 2026 *Bengaluru, India*
"DDV-GNet: High-Throughput Defect Detection for Space Manufacturing via Deep Delta Gated Networks."

Paper Presentation, IEEE GCON 2026 *IIT Guwahati, Assam, India*
"Optimizing Deep Learning for Brain Tumor Classification: A Comparative Ablation Study of Preprocessing and Augmentation Strategies."

Research Experience

Research Intern, CVPR Unit Jun. 2026 – Present

Indian Statistical Institute (ISI), Kolkata · Supervisor: Prof. Umapada Pal *Hybrid*

- Co-leading recognition of degraded Ashokan Brahmi inscriptions. Real labelled data is scarce, so we built a synthetic pipeline (WGAN-GP, 150K degraded sequences) and a SimCLR + BiLSTM-CTC recognizer (details below).
- Exploring segmentation-free restoration of degraded images, so that recognition does not depend on cleanly isolating each character first.

Research Intern Jun. 2026 – Present

The University of Salford, UK · Supervisor: Dr. Shivakumara Palaiahnakote *Remote*

- Working on text recognition in images, modelling each text line as a sequence instead of recognizing characters one at a time.
- Building a vision-language pipeline that reads sentiment in Indian editorial cartoons by combining what is drawn with what is written (details below).

Co-Founder Jan. 2026 – Present

Halo Mind Research Group *Independent*

- Co-founded, with L. Chhetri, a small group of undergraduate researchers working on efficient vision models and careful evaluation.
- Work on model design, experiments and writing. Group output includes the DSAA 2026 paper (equal-contribution first author), the ICCI 2026 paper and the Halo-Hydra patent.

Research Assistant & Team Lead Aug. 2024 – Present

Dept. of AI and Data Science, SMIT · Supervisor: Dr. Palash Ghosal *Sikkim, India*

- Lead a team of undergraduate researchers, from choosing problems and designing experiments to writing papers.
- Projects span space weather, remote sensing, defect detection and medical imaging, including my first-author TEC forecasting paper (WACV 2027, under review) and papers at ICVGIP 2026 and WIN 6.0.

Research Projects

Cross-Cycle Ionospheric TEC Forecasting (First Author) *Under Review, WACV 2027*

with Dr. Palash Ghosal | PyTorch, Deep Delta Learning, Conformal Prediction

- Built CNN-DDL, which starts from a physical persistence forecast and learns only the correction on top of it. A β-gate driven by real-time solar-wind measurements decides how much correction to apply.
- Trained on Solar Cycle 25 and tested zero-shot on Solar Cycle 24. During storms the model reached 2.30 TECU RMSE, lower than deep sequence baselines and the IRI-2020 model.
- Added split conformal prediction so every forecast comes with an interval; the intervals kept 90% empirical coverage even during extreme geomagnetic storms.

Degraded Ashokan Brahmi Script Recognition (Co-Lead) *Target: ICDAR 2027*

with Prof. Umapada Pal, ISI Kolkata | PyTorch, WGAN-GP, SimCLR, BiLSTM-CTC

- Real labelled Brahmi data is very limited, so we trained a WGAN-GP to generate 20K+ synthetic characters and applied physically motivated degradations to build a 150K-sequence training set. Team: L. Chhetri, A. Anand, G. Sarma and me.
- Recognition model: a SimCLR-pretrained ResNet-34 encoder with a BiLSTM-CTC decoder.
- Building a character error rate (CER) benchmark graded by degradation severity, to measure how much accuracy is lost going from synthetic to real inscriptions. As far as we know, it is the first such benchmark for ancient Indic scripts.

Multimodal Analysis of Indian Editorial Cartoons *Ongoing*

with Dr. Shivakumara Palaiahnakote, University of Salford | Vision-Language Models, Computer Vision, NLP

- Building a vision-language pipeline to read sentiment in editorial cartoons, where the meaning usually comes from how the drawing and the caption work together.
- Defining categories for satire, political criticism and social conflict so that commentary can be tracked systematically across cartoons.

Climate Discourse on Reddit, 2015–2025 *Independent Project*

Python, BERTopic, SBERT, Statistical Analysis

- Analysing 89K+ climate-related Reddit posts with BERTopic and SBERT embeddings to follow how topics and emotional tone changed over ten years.
- Using OLS regression and chi-square tests to check whether the rise of meta-political debate and the shift in emotional tone are statistically significant.

Patent

A. Kumar, et al. "Halo-Hydra: A Physics-Informed Generative Framework for Multimodal Visual Restoration in Underwater-Turbid and Degraded Scattering Environments." Indian Patent Office / PCT, filed (pending), 2026.

- A lightweight generative model, guided by the physics of light scattering, that restores underwater and hazy images without needing clean reference images for training. Designed with edge deployment in mind.

Academic Service & Honors

- **Peer Reviewer:** Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) 2026; IEEE GCON 2026.
- **NPTEL, IIT Kharagpur:** Introduction to Internet of Things. Elite + Silver, top 5% of candidates (87%).

Technical Skills

Programming: Python, C, Java, LaTeX

Frameworks & Libraries: PyTorch, NumPy, Pandas, scikit-learn, OpenCV

Architectures: CNNs, State-Space Models, GANs (WGAN-GP), BiLSTM-CTC, Vision-Language Models

Methods: Conformal Prediction, Self-Supervised Learning (SimCLR), Synthetic Data Generation, Topic Modelling (BERTopic, SBERT), Statistical Testing

Tools: Git, Linux